Apex PdM: Read-Aloud Script (per speaker)

> Clean lines only. Edit cues, shot IDs and graphics are in `VIDEO_SCRIPT_AND_SLIDE_BLUEPRINT.md`. Team name: **Apex PdM** (same name as the prototype).
> **Record each segment as its own file**, in a quiet room, phone 20–30 cm from your mouth. Read at a natural, energetic pace, and **leave 1 second of silence before and after** each segment. Do 2–3 takes. Say the numbers slowly.
> Each segment has a target time. Under the target is fine; over the target means trim.

Flow at a glance

#	Segment	Speaker	Target	Ends on
1	Hook	P1	0:20	"...this is what we built."
2	Problem	P2	0:30	"...go unwatched."
3	"Wait, how is it done?"	P1	0:15	"Let's open it up."
4	Hardware	P3 Adam Danish	1:00	"So... does it actually work?"
5	Live demo + data	P4 Chik Amir Ashraf	1:00	"where else could it go?"
6	Township + roadmap	P5	1:00	"...the whole township."
7	Close	P1	0:25 (incl. end card)	"Hear it before it breaks."

P1 (Host)

1. Hook (target 0:20)

> Hear that? That's a washing machine at full spin.
> Now imagine that same strain inside the air-conditioning at your mosque... ten minutes before Friday prayers.
> Machines warn us before they break. We just don't listen. So we built something that does.
> We're Team Apex PdM. And this is what we built.

3. "Wait, how is it done?" (target 0:15), say "Wait" with a beat after it

> Wait. So you might be wondering... how is it done? Is a real prototype even possible?
> Yes. And it's not a mock-up. It's running on a real washing machine, in a real, noisy home. Let's open it up.

7. Close (target 0:10 of speech)

> Machines can't tell us they're tired. So we built something that listens.
> Apex PdM. Hear it before it breaks.

P2 (Problem)

2. Problem (target 0:30)

> A township runs on machines that spin: air-conditioning, pumps, fans.
> A US Department of Energy guide found over half of facility maintenance in America is still: run it till it breaks.
> That means emergency repairs, people sweating through a prayer, and whole machines scrapped over one small part.
> Yet the same guide cites surveys where predictive maintenance eliminated 70 to 75 percent of breakdowns. The tools are costly, so most machines go unwatched.

P3 Adam Danish (Hardware)

4. Hardware (target 1:00)

> This is it. An ESP32 is the brain, this accelerometer feels the drum shake, this tiny microphone hears the motor, and this smart plug measures power.
> Feel, hear, power. Three senses on one machine, for about 170 ringgit off the shelf.
> Why a washing machine? It's a brutal test: it shakes, sloshes, and keeps changing speed.
> And our home makes it harder: an air compressor and a water pump run right beside it, the same kind of machines you'd find in a plant room.
> Here's the clever part. The processing happens on the chip. Every five seconds it turns raw sound and vibration into a set of numbers, and sends only those. No sound recordings are uploaded to the cloud. That matters in homes, and in places of worship.
> And it mounts on the outside. Nothing inside the machine is touched. So… does it actually work?

P4 Chik Amir Ashraf (Demo and data)

5. Live demo + data (target 1:00)

> Enough talking. Let's watch it work. This is our live dashboard, streaming straight from the washer.
> Vibration, sound and power, updated as it runs.
> Now watch what happens when the drum spins up. Vibration jumps more than twenty times above idle.
> And the main shaking frequency climbs to almost 20 hertz. That's the drum turning about 1,200 times a minute, measured from outside the machine.
> So far: eleven recorded wash sessions, over eight thousand rows of data.
> Then the AI learns this machine's normal, no broken machines needed. We tested it fairly: hide one wash, learn from the rest, check the hidden one. Across eight earlier washes, none was falsely flagged. And a simulated sound fault was caught seven times out of eight.
> *(Always say "simulated" and "earlier". Updated 20 Oct.)*
> And if it works on a washer… where else could it go?

P5 (Township and roadmap)

6. Township + roadmap (target 1:00)

> Why does a washing machine matter to a township? Because it's the hardest practice run.
> Think about a mosque. Rows of air-conditioning working hardest at Friday prayers, Ramadan nights and weddings. The same in temples, malls and clubhouses.
> Today, when one fails, the first people to find out are the ones sitting under it.
> Now picture a small pod on every unit, each learning its own normal and comparing itself with its identical neighbours. The odd one out gets flagged early.
> That's Planet: fix a small part instead of replacing the whole machine, and spot machines wasting power. And People: the prayer hall stays cool.
> Our roadmap. One: a pilot on about ten AC units at one township site. Two: a custom board built for the job. Three: one dashboard watching the whole township.